Stakeholder Analysis and User Stories & Use Cases

Supply Chain Performance Analysis: Shipment Delays, Cost Efficiency, Inventory Availability, and the Impact of Shortages on Patient Cases

1. Stakeholder Analysis

Stakeholder Group / Role	Type	Exact Role & Position in Project / Data	Information Needs	Pages & Questions Linked
Project Team (DEPI Team)	Internal	Four core roles - Hossam (Technical Analytics Lead), Ola (Data Quality & Review Lead), Yahia (Data Preparation & Testing Lead) and Aya (Data Dictionary & Requirements Lead): building, querying and documenting the analytical model and the dashboards.	Unified definitions, data quality rules, and reconciliation of SQL figures with Power BI.	All Pages; Q01-Q21
Lecturer & DEPI Evaluators / Executive Committee	External / Governance	Evaluation committee of the Digital Egypt Pioneers Initiative (DEPI): assesses the quality of the analysis, methodological transparency, and the ability of the dashboard to support decisions.	Project files on GitHub and the LMS, before/after cleaning documentation, and every recommendation linked to a sample and an exact measure, without unsupported causal claims.	All Deliverables
Chief Operating Officer (COO) & Executive Management	Internal Decision Owner	Senior and executive management of the health network: follows overall performance, the general service level and total cost.	Number and value of shipments by year and product group, headline hospital figures (PO On-Time Rate, Stockout Rate, Unmet Demand EGP), data completeness, and the decision summary.	Page 1 Executive Overview; Q01-Q03, Q21
Global Supply Chain & International Freight Manager	Internal Decision Owner	Manager of international logistics supply chains: directly follows shipments from international vendors (such as IDA Foundation, Novartis, Amstelfarma and Imres) across countries (such as Nigeria, Haiti, Cote d'Ivoire and DRC).	Schedule status, Lead Time from the purchase order date to delivery (by country), customs clearance days, shelf life at delivery and receiving inspection results.	Page 2 Delivery & Cycle Time; Q04-Q07
Global Procurement & Cost Controller Manager	Internal Decision Owner	Manager of external procurement and cost control: evaluates vendor performance and cost relative to value.	Vendor value share, value and unit price by product group and molecule, freight cost and Freight/Value, Insurance/Value, and freight against weight.	Page 3 Suppliers & Cost; Q08-Q11
Hospital Local Procurement & Central Warehouse Manager	Internal Decision Owner	Manager of local procurement and central warehouses for the hospitals: directly follows the purchase_orders table and the actual commitment of the eight suppliers (S01 to S08) and the regional distribution center SCMS - RDC (S09).	Actual on-time rate against the historical on-time rate, PO fill ratio, distribution of PO delay days, purchase order status, and actual lead time against item baseline.	Page 4 Purchase Orders & Inventory; Q12-Q14
Hospital Network Inventory Officers	Operational Users	Inventory officers and pharmacists in the four hospitals: H-A (Teaching), H-B (General), H-C (District) and H-D (Maternity & Children's).	Stockout Rate by hospital and criticality with its weekly trend, estimated value of unmet demand in EGP, and estimated value of wastage.	Page 4 Purchase Orders & Inventory; Q15-Q16

Stakeholder Group / Role	Type	Exact Role & Position in Project / Data	Information Needs	Pages & Questions Linked
Chief Medical Officer & Clinical Quality Reviewer	Internal Decision Owner	Manager of medical affairs and clinical quality in the hospitals: assesses the clinical impact of shortages of medicines and supplies.	Number of patient cases by treatment_status, treatment delay hours, outcome, ICU admission, complication and length of stay, the link between item stockouts and delayed treatment, and diagnoses and hospitals with higher shortage-type shares.	Page 5 Case Impact & Decision Quality; Q17-Q20
All Dashboard Users	Internal / External Users	All dashboard users, including department managers and decision leaders: review the decision summary.	Summary of the top improvement opportunities supported by the numbers, sample sizes, and the stability of results after data cleaning.	Page 5 Decision Summary; Q21

2. User Stories

Format: As [stakeholder], I want to know [question], so that [decision].

Page 1 - Executive Overview

ID	User Story	Source Columns	Visuals
Q01	As the Chief Operating Officer (COO) & Executive Management, I want to know how many shipments were recorded, their total value, and how the value splits by year and by product group, so that the size and composition of the shipment portfolio is known.	id; line item value; delivered to client date; product group	KPI cards; column chart by year; donut by product group
Q02	As the Chief Operating Officer (COO) & Executive Management, I want to know the headline hospital figures: purchase orders, inventory days and cases, with PO On-Time Rate, Stockout Rate and Unmet Demand EGP, so that the hospital layer can be read next to the shipments on one page.	purchase_orders (po_id, expected_delivery_date, received_date); inventory_daily (unmet_demand_qty); items.unit_cost_egp; patient_cases.case_id	KPI cards
Q03	As the Chief Operating Officer (COO) & Executive Management, I want to know how complete the key fields of both families are after cleaning, so that the reliability of every other page is visible.	Key fields of the shipments and hospital tables	Bar chart of completeness by field

Page 2 - Delivery & Cycle Time

ID	User Story	Source Columns	Visuals
Q04	As the Global Supply Chain & International Freight Manager, I want to know how many shipments were delivered on schedule, earlier or later than scheduled, so that delivery performance is stated with its denominator.	scheduled delivery date; delivered to client date	Stacked bar of Schedule Status
Q05	As the Global Supply Chain & International Freight Manager, I want to know the distribution of lead time from PO to delivery recorded, and the median lead time by country, so that cycle time is reported with sample size, for countries with at least 5 shipments.	po sent to vendor date; delivery recorded date; country	Histogram; bar of median by country
Q06	As the Global Supply Chain & International Freight Manager, I want to know whether customs clearance days relate to lead time, so that the share of cycle time spent in customs is understood without implying causation.	customs_clearance_days; Lead Time Days	Scatter plot
Q07	As the Global Supply Chain & International Freight Manager, I want to know how much shelf life remained at delivery, which shipments arrived already expired, and the receiving inspection results, so that expiry risk at delivery is identified from the records.	expiry_date; delivered to client date; receiving_inspection_result; id	Histogram; donut; table of expired-on-arrival shipments

Page 3 - Suppliers & Cost

ID	User Story	Source Columns	Visuals
Q08	As the Global Procurement & Cost Controller Manager, I want to know which vendors account for the largest share of shipment value, so that vendor concentration is known; vendors with fewer than 3 shipments are grouped as Other.	vendor; line item value	Pareto chart
Q09	As the Global Procurement & Cost Controller Manager, I want to know how value and unit price differ by product group and molecule or test type, so that price and value drivers are compared within the same molecule.	product group; molecule/test type; unit price; line item value	Bar chart of value; bar chart of unit price
Q10	As the Global Procurement & Cost Controller Manager, I want to know the total freight and insurance cost and the Freight/Value ratio by country and by product group, so that cost efficiency is computed from valid numeric records only.	freight cost (usd); line item insurance (usd); line item value; country; product group	KPI cards; two bar charts
Q11	As the Global Procurement & Cost Controller Manager, I want to know whether freight cost grows with weight, the Freight per Kg, and where the outliers are, so that outliers are flagged for review instead of being deleted.	weight (kilograms); freight cost (usd); id	Scatter plot; outlier table

Page 4 - Purchase Orders & Inventory

ID	User Story	Source Columns	Visuals
Q12	As the Hospital Local Procurement & Central Warehouse Manager, I want to know each supplier's actual on-time rate against its historical rate, and its PO fill ratio, so that the difference is read with the caveat that period and measurement may differ.	purchase_orders (supplier_id, expected_delivery_date, received_date, qty_ordered, qty_received); suppliers.historical_on_time_rate	Clustered bar; bar of fill ratio
Q13	As the Hospital Local Procurement & Central Warehouse Manager, I want to know how many days received purchase orders are late and how orders split by status, so that the PO delay distribution and the status mix are known.	expected_delivery_date; received_date; po_status	Histogram; donut
Q14	As the Hospital Local Procurement & Central Warehouse Manager, I want to know how actual order-to-receipt time compares with the item baseline lead time, so that categories where actual lead time exceeds the baseline are identified.	order_date; received_date; items.baseline_lead_time_days; items.category	Bar chart of difference by item category
Q15	As the Hospital Network Inventory Officers, I want to know where the Stockout Rate is concentrated by hospital and criticality, and how it moves week by week, so that the rate and the count are shown, not the quantity only.	inventory_daily (unmet_demand_qty, hospital_id, item_id, date); items.criticality	Heatmap; line chart
Q16	As the Hospital Network Inventory Officers, I want to know the estimated value in EGP of unmet demand by item and of wastage by item category and cold-chain flag, so that values are described as estimates and not as realized losses.	unmet_demand_qty; expired_writeoff_qty; damaged_lost_qty; items.unit_cost_egp; items.category; items.cold_chain	Bar chart (top 10 items); stacked bar

Page 5 - Case Impact & Decision Quality

ID	User Story	Source Columns	Visuals
Q17	As the Chief Medical Officer & Clinical Quality Reviewer, I want to know how many cases fall in each treatment status and how treatment delay hours are distributed, so that denominators and missing data are visible.	treatment_status; treatment_delay_hours; case_id	Bar chart; box plot

ID	User Story	Source Columns	Visuals
Q18	As the Chief Medical Officer & Clinical Quality Reviewer, I want to know how outcome, ICU admission, complication and length of stay differ by treatment status, and the death share within each severity level, so that the comparison is descriptive, with group sizes shown and no causal claim.	outcome; icu_admission; complication; length_of_stay_days; severity; treatment_status	100% stacked bar; table; matrix
Q19	As the Chief Medical Officer & Clinical Quality Reviewer, I want to know how each item's Stockout Rate compares with the share of its cases delayed because stock was unavailable, so that the descriptive link between item stockouts and delayed treatment is visible.	patient_cases.key_drug_item_id; treatment_status; inventory_daily.unmet_demand_qty; item_id	Scatter plot by item
Q20	As the Chief Medical Officer & Clinical Quality Reviewer, I want to know which diagnoses and hospitals have higher shares of shortage-type cases, so that affected diagnoses and hospitals are named, above a minimum group size.	diagnosis; hospital_id; treatment_status; case_id	Two bar charts
Q21	As a dashboard user, I want to know the top improvement opportunities supported by the numbers and how stable the results are after cleaning, so that every recommendation is linked to a measure, a sample size and stated limits.	KPI results; flags; reconciliation log	Decision summary table

3. Use Cases

ID	Use Case	Actor	Precondition	Main Flow	Rules and Outcome
UC-01	Review the Executive Overview	Chief Operating Officer (COO) & Executive Management	Cleaned models are refreshed.	1. Open Page 1. 2. Read the shipment and hospital KPI cards. 3. Read shipment value by year and by product group. 4. Read data completeness by field.	No recommendation is shown without a KPI and a sample size; shipment figures stay separate from hospital figures.
UC-02	Review delivery and cycle time	Global Supply Chain & International Freight Manager	Delay Days, Lead Time Days, Customs Clearance Days and Shelf Life fields exist with valid-date flags.	1. Open Page 2. 2. Read Schedule Status. 3. Compare lead time by country. 4. Read customs against lead time. 5. Review shelf life, inspection results and the expired-on-arrival table.	Invalid dates are not used; n is always shown; correlation is not treated as cause.
UC-03	Review vendor value and cost	Global Procurement & Cost Controller Manager	Valid numeric freight, weight and value fields exist.	1. Open Page 3. 2. Read the vendor Pareto. 3. Compare value and unit price by product group and molecule. 4. Read Freight/Value by country and product group. 5. Read freight against weight and the outlier table.	Ratios are calculated from valid sums; groups below the minimum size appear as Other.
UC-04	Review purchase orders, stockouts and wastage	Hospital Local Procurement & Central Warehouse Manager; Hospital Network Inventory Officers	Hospital-layer tables are cleaned and keys are validated.	1. Open Page 4. 2. Compare actual and historical on-time rates. 3. Read PO delay and status. 4. Read lead time against baseline. 5. Read the Stockout heatmap and trend. 6. Read unmet demand and wastage in EGP.	Hospital data is separate and synthetic; EGP values are estimates and not proven losses.
UC-05	Review case impact and decision quality	Chief Medical Officer & Clinical Quality Reviewer	Patient-case fields are standardized and key_drug_item_id is linked to items.	1. Open Page 5. 2. Read cases and delay hours by treatment status. 3. Compare outcomes within each severity level. 4. Read item stockout against delayed-treatment share. 5. Read diagnoses and hospitals. 6. Read the decision summary and its limits.	No causal medical claim; group sizes and missing values are shown.